Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS Computer Science, 2.5+ years) with deep expertise in LLM fine-tuning adjacent work, RAG systems, agentic workflows, and LLM evaluation pipelines. Delivered ~35% retrieval precision gain, >30% hallucination reduction, and 25% agent stability improvement in production healthcare environments. Expert in Python, PyTorch, LangChain, LlamaIndex, and vector databases. Strong communicator - proven ability to explain complex AI systems to technical and non-technical audiences. Eligible for visa sponsorship.

CORE COMPETENCIES

LLM Fine-tuning (PyTorch / HuggingFace) • RAG Systems & Advanced Retrieval • Agentic Workflows & Orchestration • LLM Evaluation Pipelines • LangChain / LlamaIndex • Vector Databases & Embeddings • Python / PyTorch / FastAPI / Docker • Prompt Engineering & Structured Outputs • Customer Onboarding (Technical) • Pre-Sales Technical Guidance

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS

- Built **RAG systems** for clinical knowledge retrieval with recursive semantic chunking, transformer embeddings, and reranking; improved contextual retrieval precision **~35%**, reduced irrelevant retrieval **>30%**; grounding alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules; improved agent stability **~25%**, reduced hallucinations **>30%**.
- Built **LLM evaluation pipelines**: precision/recall tracking, hallucination measurement, grounding alignment scoring, audit-ready evaluation reports for stakeholders.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient risk; improved recall **15-20%** on high-risk categories, precision >90%.
- Packaged inference as **FastAPI/Flask services** with Docker; reduced post-deploy defects **~30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer - Applied NLP, Video Analytics

- Built **transformer-based domain adaptation** (hierarchical abstractive summarization + timestamp alignment) across 5,000+ sessions; **60-70%** review time reduction, **~85%** highlight alignment with human curation.
- RAG-grounded document Q&A: reduced hallucinations **~30%**, raised contextual accuracy **>85%**. Flask API.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

- SQL/T-SQL optimization (~20% latency reduction); Jenkins CI/CD (>30% deployment error reduction); Grafana monitoring.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with **schema-validated JSON contracts** between agents to prevent hallucination cascades. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN-based duplicate detection; explainable risk scoring with audit-ready reasoning traces. Demonstrates production-grade agentic LLM deployment with evaluation rigor.

Agentic Pixel Character Synthesis - LLM + Generative AI Orchestration

LLM-based agent orchestrator decomposes high-level prompts into Stable Diffusion + LoRA fine-tuning + ControlNet generation tasks. PyTorch, HuggingFace Diffusers, FastAPI, Docker. Demonstrates hands-on fine-tuning (LoRA), agentic orchestration, and multimodal generative pipeline design.

Dream Decoder - Multimodal AI with Intermediate Prompt Transformation

FastAPI + React app using Perplexity Sonar + Replicate. Introduced structured prompt transformation layers; **~30% contextual alignment improvement**, **~25-30% first-pass image success** over naive direct-prompt orchestration. Demonstrates advanced RAG + generation pipeline engineering.

Driver Drowsiness Detection - YOLOv8 Real-Time System

Unified YOLOv8 replacing two-stage CNN; **~30%** inference latency reduction. Sliding-window confidence aggregation: **~25%** false positive reduction. OpenCV real-time capture pipeline.

EDUCATION

Master of Science, Computer Science - ***Kent State University, Ohio*** *Aug 2023 - May 2025*

SKILLS

Languages: Python, TypeScript, SQL, C++ (Arduino)

LLM & GenAI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, evaluation pipelines, vector search, embeddings, fine-tuning (LoRA, HuggingFace)

ML & Vision: PyTorch, scikit-learn, XGBoost, HuggingFace Transformers, YOLOv8, Stable Diffusion, ControlNet

Infra: Docker, FastAPI, Flask, Jenkins, Grafana, REST APIs, CI/CD

Work Authorization: F-1 OPT (US-authorized now); visa sponsorship eligible